

Week 4: Continuum fluid dynamics

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Abstract: This week we learn to integrate a differential equation in time and to know when the result can be trusted. This week the unknown is a field: two-dimensional incompressible flow, written through vorticity and a streamfunction and differentiated with Fourier multipliers. Once space is discretized, the flow is a large system of ordinary differential equations that the integrators of week 2 advance. We build forward Euler, the explicit midpoint rule, and fourth-order Runge–Kutta behind one interface and test them first on a wave carried along a periodic line, where every answer is exact: the step at which each blows up, and the power with which each error falls. We then build the flow solver on them, read their stability regions against the solver’s spectrum, predict the largest stable step for diffusion and for advection and measure both by running past them, verify the nonlinear solver on a random flow, and choose a grid and time step by Richardson estimation.

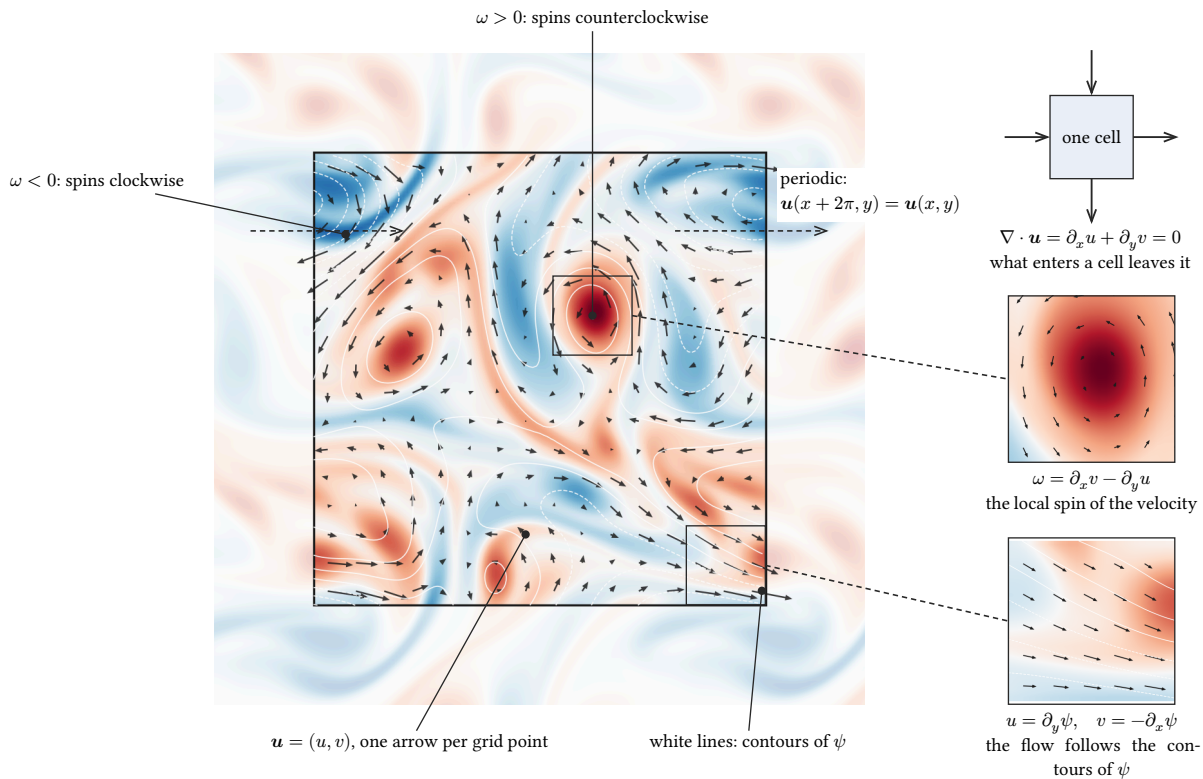
Resources:

- Trefethen, *Finite Difference and Spectral Methods for Ordinary and Partial Differential Equations* [1], free.

The problem: a fluid as a field

Before reading on, watch 3Blue1Brown, *Divergence and curl*, a 16-minute video: a velocity field as arrows, divergence as what enters against what leaves, curl as local spin, all drawn on a fluid.

Atoms in a cup of water number 10^{25} , so a fluid is described instead by what is at each *point*: a **velocity field** $\mathbf{u}(x, t)$, an arrow at every point x and time t . The grid of points stays put while the fluid passes through it: the *Eulerian* picture. Our fluid is *incompressible*: what enters a cell leaves it. It lives on a *periodic* square: what leaves the right edge re-enters on the left, and there are no walls. It has a small viscosity ν . We let a random flow decay and watch nonlinear transport change the sizes of its vortices.



The objects of the problem, drawn from the random case of this week at $t = 5$. Colour is the vorticity ω , the local spin defined below, red positive and blue negative; the box continues faded beyond its edges because each edge is glued to the opposite one. The tiles enlarge the two framed windows.

The field evolves under two conservation laws.

Mass. In an **incompressible** fluid the density is constant, so the fluid entering a cell must equal the fluid leaving it, in every cell, at every instant; with the components $\mathbf{u} = (u, v)$,

$$\nabla \cdot \mathbf{u} = \partial_x u + \partial_y v = 0. \quad (1)$$

Momentum. Newton's law for a parcel of fluid, the incompressible Navier–Stokes equation, reads

$$\underbrace{\partial_t \mathbf{u}}_{\text{local change}} + \underbrace{(\mathbf{u} \cdot \nabla) \mathbf{u}}_{\text{advection}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \nabla^2 \mathbf{u}}_{\text{viscosity}}, \quad (2)$$

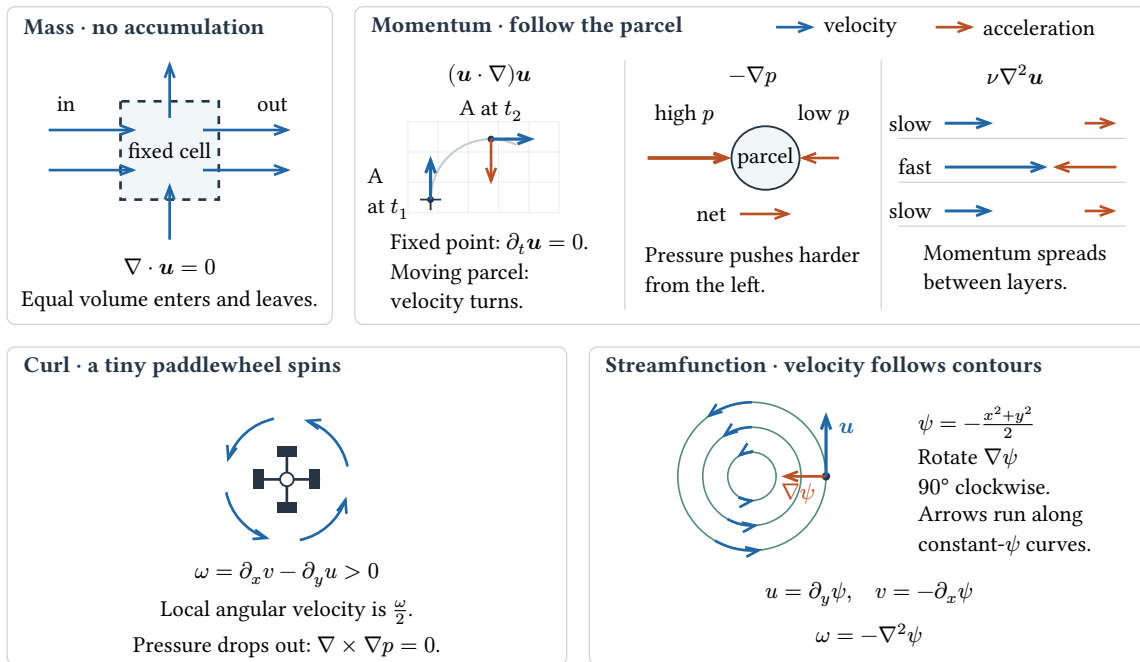
the acceleration of a moving parcel on the left, pressure pushing from high to low and viscosity spreading momentum between neighbouring layers on the right; p is pressure divided by the constant mass density and ν the **kinematic viscosity**. Pressure adjusts to keep Equation 1 true, which makes Equation 2 awkward to integrate directly.

Vorticity and streamfunction. In two dimensions the **vorticity** $\omega = \partial_x v - \partial_y u$ is a scalar, the local spin of the fluid. Taking the curl of Equation 2 kills ∇p , because the curl of a gradient is zero, and leaves one scalar equation:

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \nabla^2 \omega \quad (3)$$

the **vorticity equation**. Vorticity is carried along by the flow, the middle term, and diffused by viscosity, the right side. To advance this equation, the solver must recover $\mathbf{u}$ from ω at every step.

We recover this velocity from a *streamfunction* ψ , with $u = \partial_y \psi$, $v = -\partial_x \psi$, a field that satisfies Equation 1 identically, since $\partial_x \partial_y \psi - \partial_y \partial_x \psi = 0$. Its vorticity is $\omega = -\nabla^2 \psi$. So the recovery is a Poisson equation: given ω , solve $-\nabla^2 \psi = \omega$, then differentiate. A velocity built this way is divergence-free to rounding error, whatever the solver does to ω .



From conservation laws to the solver. These local sketches show what each equation means physically. Taking curl removes pressure from the momentum equation; solving for ψ recovers a velocity that obeys mass conservation. The contours show the flow direction at one instant.

Given ν , an initial ω and the periodic box, Equation 3 with the streamfunction recovery determines the flow for all time. Week 2 faced the same kind of problem with a few atoms: Newton's law gave the rate of change of every position and velocity, and an integrator advanced them step by step. Here the unknown is the field ω at N^2 grid points. Once the spatial derivatives are discretized, Equation 3 becomes a system of N^2 ordinary differential equations, $d\omega/dt = F(\omega)$, the *method of lines*, and the integrators of week 2 apply to it unchanged. What is new is the size of the system and the spread of its rates: the slowest large-scale motion and the fastest decay of the shortest resolved wave differ by orders of magnitude, and that spread decides which integrator and which step are allowed.

KEY CONCEPTS

- A *velocity field* assigns a velocity to each fixed point in space, the Eulerian description.

- *Incompressibility* requires $\nabla \cdot \mathbf{u} = 0$: fluid entering a cell balances fluid leaving it.
- *Vorticity* measures local rotation; taking the curl removes pressure from the evolution equation.
- The *streamfunction* recovers a divergence-free velocity from vorticity through a Poisson equation.
- *Advection* carries vorticity with the flow; *viscosity* diffuses it between neighbouring regions.
- The *method of lines* discretizes space first, leaving a large system of ordinary differential equations.

Part 1: Integrators on a line

Goal: build forward Euler, the explicit midpoint rule, and classical fourth-order Runge–Kutta behind one interface, and test them where every answer is exact: a wave carried and diffused along a periodic line. Verify that RK4 blows up once its shortest wave leaves the stability region, in stripes of that wave, and that the three errors fall with slopes 1, 2, and 4.

① DO

Work in the `AMAT5315-2026Fall-Exercise` repository from weeks 1 to 3. The agent runs at its root and receives the blue panels; you run the grey panels in a second terminal. In that terminal, create the week’s folder and enter it:

```
> mkdir -p week4/evidence && cd week4
```

Set the working rules:

Three rules for this week's work in ``week4/``: after each of my prompts, commit the work before you reply; when anything is uncertain, ask me instead of guessing; save every script you write for a run, a comparison, or a plot under ``week4/scripts/`` and run it from there.

Then the integrators and the toy they are tested on; a **trait** is Rust’s word for one interface shared by several types:

Build a Rust library in ``week4/`` with one ``Integrator`` trait shared by forward Euler, the explicit midpoint rule, and classical fourth-order Runge-Kutta: each advances a state vector by one step of a given size from a given rate function. Add the rate function of the advection-diffusion equation $u_t + c u_x = \nu u_{xx}$ on a periodic $[0, 2\pi]$ grid of n points with Fourier derivatives, and a second one with centred finite differences. The Nyquist mode $k = -n/2$ does not travel: its first derivative is zero. Test every integrator against the exact solution of a single wave.

Read on while the agent builds.

② UNDERSTAND

Ordinary and partial differential equations. Week 2 followed atoms: the state was their positions and velocities, and Newton’s law gave the rate at which each changes. Equation 3 is the same kind of law for the field ω . Both are *differential equations*, stated formally as follows.

Definition. Let $y : [0, T] \rightarrow \mathbb{R}^m$ be an unknown function of time, the **state vector**. An *ordinary differential equation* (ODE) is a relation between the state and its time derivative,

$$\frac{dy}{dt} = f(t, y(t)), \quad 0 < t \leq T, \quad (4)$$

with $f : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ given. A *solution* is a differentiable y that satisfies Equation 4 at every t . The equation fixes the direction of motion, not the position: Equation 4 together with an *initial condition* $y(0) = y_0$ is an *initial value problem*, and for f Lipschitz in y it has exactly one solution [1, §1.1]. A law with higher derivatives reduces to Equation 4 by adding them to the state, as Newton’s $m d^2x/dt^2 = F$ becomes (x, v) with $dx/dt = v$, $dv/dt = F/m$. When the unknown $u(x, t)$ depends on space $x \in \Omega \subset \mathbb{R}^d$ as well, and the law involves its spatial derivatives $\partial_{x_i} u$, $\partial_{x_i} \partial_{x_j} u$, it is a *partial differential equation* (PDE),

$$\partial_t u = G(u, \nabla u, \nabla^2 u, \dots), \quad x \in \Omega, \quad 0 < t \leq T, \quad (5)$$

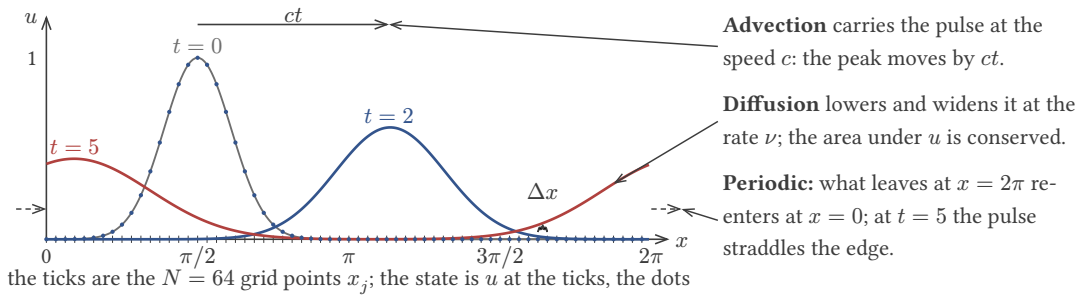
closed by the initial condition $u(x, 0)$ on Ω and a *boundary condition* on $\partial\Omega$; here Ω is a periodic square and the boundary condition is periodicity.

Here $\partial_t u$ is the rate of change at a fixed point and ∇u collects the slopes along each space direction; heat, waves, diffusion, electromagnetism, and Equation 3 are all of the form Equation 5. To solve Equation 4 numerically is to advance the state in small steps Δt , as the integrators of week 2 do. To solve Equation 5, this week discretizes space first, which turns Equation 5 into a system of the form Equation 4 with large m : the method of lines. This Part does so on the simplest field equation that has both of the fluid's ingredients, so that every step can be checked against a formula.

The advection–diffusion line. This Part's unknown is a scalar $u(x, t)$ on the periodic interval $[0, 2\pi)$, carried at a constant speed c and diffused by a viscosity ν , the **advection–diffusion equation**:

$$\partial_t u + c \partial_x u = \nu \partial_x^2 u \quad (6)$$

It is Equation 3 with three things removed: the speed that carries u is a constant instead of a field recovered from u , so there is no product and no Poisson solve, and there is one dimension instead of two. Part 2 adds those three; the grid, the derivatives and the integrators are built here and reused unchanged. The state of this Part is a **Gaussian pulse**, $u(x, 0) = e^{-(x-\pi/2)^2/(2\sigma^2)}$ of standard deviation σ centred at $x = \pi/2$, summed over its periodic images so that it is periodic on the box, and sampled at the grid points.



The line and its pulse. The exact solution of Equation 6 for a Gaussian pulse of standard deviation 0.35 started at $x = \pi/2$, at $\nu = 0.05$, $c = 1$, $N = 64$: the state of Part 1 is this curve sampled at the ticks.

Spatial derivatives on the periodic grid. Sample the interval at $x_j = j\Delta x$, $\Delta x = 2\pi/N$, $j = 0, \dots, N-1$, a **periodic grid**: the endpoint 2π is the same place as 0. The N samples determine a finite Fourier sum

$$u(x_j) = \sum_{k=-N/2}^{N/2-1} \hat{u}_k e^{ikx_j}, \quad \hat{u}_k = \frac{1}{N} \sum_{j=0}^{N-1} u(x_j) e^{-ikx_j}, \quad (7)$$

whose integer **wavenumbers** k count oscillations around the box; the fast Fourier transform (FFT) computes either sum in $N \log N$ operations and stores the coefficients in the order $0, 1, \dots, N/2-1, -N/2, \dots, -1$. A derivative of one wave brings down its wavenumber, $\partial_x e^{ikx} = ik e^{ikx}$, so a **Fourier derivative** is computed by multiplying each $\hat{u}_k$ by ik . For the second derivative, the *Fourier multiplier* is $-k^2$. The mode $k = -N/2$, the **Nyquist mode**, has no partner $+N/2$ on the grid, so its odd derivative is set to zero. The alternative is local: the **centred finite differences**

$$\partial_x u(x_j) \approx \frac{u_{j+1} - u_{j-1}}{2\Delta x}, \quad \partial_x^2 u(x_j) \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}, \quad (8)$$

with neighbours wrapping around the ends, whose errors fall as Δx^2 . With either choice the state is the vector $u = (u_0, \dots, u_{N-1})$ and Equation 6 becomes $du/dt = F(u)$ with F the **rate function**, an instance of Equation 4 with $m = N$: space is discretized first and time is left continuous, the *method of lines*.

The spectrum of the line. For a Fourier wave other than the Nyquist mode, ∂_x gives ik and ∂_x^2 gives $-k^2$, so

$$\frac{d\hat{u}_k}{dt} = \lambda_k \hat{u}_k, \quad \lambda_k = -\nu k^2 - ick, \quad \hat{u}_k(t) = \hat{u}_k(0) e^{\lambda_k t}. \quad (9)$$

the **exact solution of a single wave**. The multipliers are exact on the grid's waves. The **modes** evolve independently: each non-Nyquist mode travels at speed c and decays at the rate νk^2 , the short waves fastest. The numbers λ_k are the *spectrum* of F : diffusion contributes the negative real part and advection the imaginary part. The Nyquist mode has zero first derivative, so $\lambda_{-N/2} = -\nu(N/2)^2$: it decays but does not travel. It has the fastest diffusive decay; the largest advection frequency occurs at $|k| = N/2 - 1$. Centred differences replace ik by $i \sin(k\Delta x)/\Delta x$, smaller for short waves, so with them short waves travel too slowly and a sharp pulse trails a wake. A linear equation splits into modes that evolve alone; the spectrum says how fast each moves and decays, and its farthest point decides the time step.

Three explicit integrators. A **time integrator** advances the state from t_n to $t_n + h$, with $h = \Delta t$ the **time step**. This week's three are *explicit*: each evaluates F only at states already computed within the step. With $u_n \approx u(t_n)$,

$$\begin{aligned} \text{Euler: } u_{n+1} &= u_n + hF(u_n), \\ \text{midpoint (rk2): } k_1 &= F(u_n), \quad k_2 = F\left(u_n + \frac{h}{2}k_1\right), \quad u_{n+1} = u_n + hk_2, \\ \text{RK4: } k_1 &= F(u_n), \quad k_2 = F\left(u_n + \frac{h}{2}k_1\right), \quad k_3 = F\left(u_n + \frac{h}{2}k_2\right), \quad k_4 = F(u_n + hk_3), \\ u_{n+1} &= u_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \tag{10}$$

Forward Euler is week 2's first stepper. The **midpoint rule** takes a trial half step and uses the rate found there; **RK4**, the classical fourth-order Runge–Kutta, blends four rates with the weights 1, 2, 2, 1. Each **stage** evaluates F at its own trial state, and the weights are not a matter of taste: **equal weights**, 1, 1, 1, 1 over 4, give a method that still runs and still converges, but only as fast as the midpoint rule; its last two stages buy nothing.

The stability function. Every linear property of an integrator shows on the **test equation** $dy/dt = \lambda y$, since a linear system splits into such modes. One step multiplies y by the **stability function** $R(z)$ of the method, with $z = \lambda h$, so $|R(z)|$ is the **growth factor per step**:

$$R_{\text{Euler}}(z) = 1 + z, \quad R_{\text{midpoint}}(z) = 1 + z + \frac{z^2}{2}, \quad R_{\text{RK4}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \tag{11}$$

each the Taylor polynomial of the exact factor e^z cut after the term of its *accuracy order* $p = 1, 2, 4$. The error of one step is proportional to h^{p+1} , and the error at a fixed final time to h^p . A negative real λ means decay, an imaginary λ means oscillation, and neither grows. The step is **stable** for a mode when $|R(z)| \leq 1$, so that the numerical mode does not grow either; the set of such z in the complex plane is the method's *stability region*.

Euler's region is the disc of radius 1 about -1 . It contains no point of the imaginary axis except 0, so every oscillating mode grows a little each step: that growth was the energy drift of week 2. The midpoint rule reaches the same real limit, -2 , and also misses the imaginary axis. RK4's region reaches -2.785 on the real axis and $\pm 2.83i$ on the imaginary one. Velocity-Verlet covers only the imaginary segment, exactly the modes atoms have, and none of the decaying ones: the right tool for week 2 and the wrong one for a diffusing wave. An integrator suits a problem when the problem's modes fit inside its stability region; the accuracy order says nothing about that.

The largest stable step of the line. By Equation 9 the mode k sits at $z = \lambda_k h = -\nu k^2 h - ickh$, on a parabola through the origin; as h grows the whole set stretches away from it, and the first point to leave the region is the shortest wave that still travels, $|k| = 31$, since the Nyquist wave's odd derivative was dropped and it sits on the real axis. The largest stable step is the h at which that point touches the boundary, found by bisection on $|R(\lambda_k h)| \leq 1$ over all modes. At $\nu = 0.05$, $N = 64$, the parabola leaves through the curved part of RK4's region near its real crossing: the exact limit is $h = 0.0494$, while the real-axis estimate alone, $\nu k^2 h \leq 2.785$, would say 0.0544; the advective part pushes the tip out where the region is narrow. The axis estimates are what Part 3 can afford on the fluid, whose spectrum is not known exactly; here the map shows both the estimate and the truth. Past the boundary the offending mode multiplies by $|R(z)| > 1$ every step, starting from round-off, so the growth needs many steps to show: the boundary is sharp in $|R|$, the time to see it is not.

Accuracy order. Inside the region the question turns from whether the error grows to how fast it shrinks. On log-log axes a power law for the **maximum error** over the grid, $\text{error} \propto h^q$, has slope q . A least-squares fit in log-log space gives the **fitted slope**. Alternatively, estimate q from two steps h_1, h_2 with errors $\text{err}_1, \text{err}_2$: $q = \log(\text{err}_1/\text{err}_2)/\log(h_1/h_2)$. The slope is the method's accuracy order p once the steps are small enough for the leading term h^p to dominate, the method's *asymptotic range*; every run of the series must also be stable, or the error measures growth rather than accuracy. Stability asks whether the error grows; accuracy asks how fast it shrinks with the step. Both are properties of the pair, integrator and spectrum.

- Read Trefethen [1], chapter 1, sections 1.7 to 1.9. Find the stability regions of the Runge–Kutta family and the statement that no explicit method has an unbounded stability region.

KEY CONCEPTS

- A *differential equation* relates a state to its rate of change, [Equation 4](#); with an *initial condition* it has exactly one solution. It is *ordinary* when the state depends on time alone, *partial* when it also depends on space, [Equation 5](#).
- The *method of lines* samples space on a periodic grid and leaves time continuous; *Fourier multipliers* differentiate the grid's waves exactly, *centred differences* carry a Δx^2 error and slow the short waves.
- A linear equation's *spectrum* is one λ_k per mode, [Equation 9](#): diffusion on the negative real axis, advection on the imaginary axis.
- A *time integrator* advances the state by one step, [Equation 10](#); its *stability function* $R(z)$, $z = \lambda h$, is what one step does to a mode, and the *stability region* $|R(z)| \leq 1$ decides which steps are allowed.
- RK4 has both a real and an imaginary stability interval, which diffusion and advection require; Euler and midpoint have no imaginary one, *velocity-Verlet* only that one.
- The *accuracy order* p is the power of h the error follows, read as a log-log slope inside the asymptotic range.
- **The check:** the measured growth map against the analytic boundary and a pulse on either side of the limit, then the pulse profiles and the four error slopes.

3 VERIFY

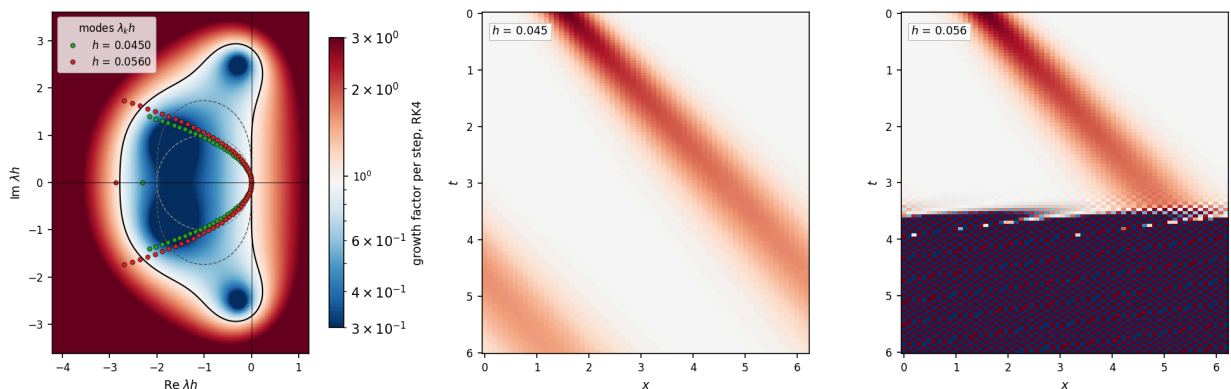
(1) RK4 blows up where the stability function says, in the shortest wave.

Draw the complex plane of $z = \lambda h$ coloured by the growth factor per step measured by advancing $y' = \lambda y$ one step of $h = 1$ with the RK4 of the library in `'week4/'`, on a log colour scale, with the curves where each of the three integrators' stability function has modulus one, and the modes of the line at $\nu = 0.05$, $n = 64$, $c = 1$ as dots for the steps 0.045 and 0.056, as `'week4/evidence/line-stability.png'`.

Add two panels to `'week4/evidence/line-stability.png'`: u as a colour image, x across and t downward, for a Gaussian pulse of standard deviation 0.35 centred at $x = \pi/2$, periodic on the box, integrated by the library's RK4 to $t = 6$ at $\nu = 0.05$, $n = 64$, $c = 1$, once at $dt = 0.045$ and once at $dt = 0.056$.

Open `evidence/line-stability.png`.

Expected output: the figure below. Required: the coloured edge sits on the black curve, and the curve crosses the axes at -2.785 and $\pm 2.83i$; every dot at 0.045 lies inside it and the outermost dots at 0.056 outside, the tip of the parabola touching the curve at $h = 0.049$; the run at 0.045 is one clean diagonal stripe to $t = 6$; the run at 0.056 follows the same stripe and then erupts, before $t = 6$, into stripes two grid points wide.



The stability map, measured, and a pulse on either side of the limit. Left: colour is the growth factor per step measured by running RK4 on $y' = zy$; the black curve is $|R_{RK4}(z)| = 1$, the dashed ones Euler's and midpoint's. The dots are the line's modes $\lambda_k h$ at the two steps, on the parabola of [Equation 9](#). Middle and right: $u(x, t)$ at $h = 0.045$ and 0.056 , below and above the exact limit 0.0494; the pulse circles the box once in 2π time units.

Explanation. The map is the theory checked by brute force: the stepper itself, not its formula, decides where growth begins, and the two agree everywhere. The instability begins with the wave $|k| = 31$, at the outer end of the parabola. This mode lies outside the stability region at $h = 0.056$ and is the first to become unstable as the

step increases, so the picture fills with the pattern of that one wave, alternating from one grid point to the next. Nothing in the pulse drives it; round-off seeds it and $|R| = 1.85$ per step amplifies it, which is why it stays invisible for three time units and then takes over within a few steps. Between 0.049 and 0.056 the same wave grows more slowly and surfaces later; at 0.0535 it takes five time units, and the growth is visible by $t = 6$ only from 0.0535 upward. A tip that leaves the curve elsewhere than at 0.049 means the spectrum is not what Equation 9 says: a missing factor in a multiplier, or a Nyquist wave that still travels.

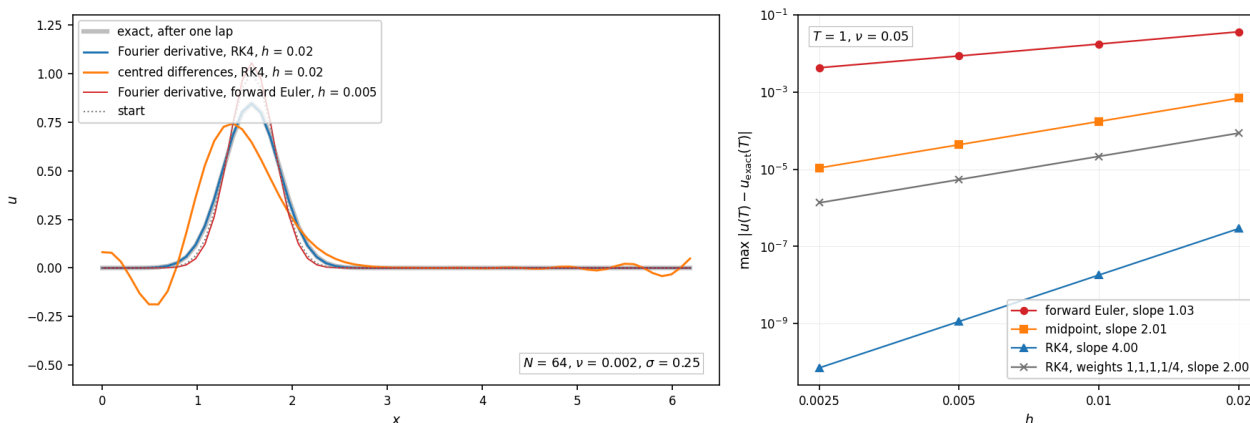
(2) Spatial and temporal errors in pulse propagation.

Use the library in `week4/` to evolve a periodic Gaussian pulse centred at $x = \pi/2$ with standard deviation 0.25. Use $n = 64$, $c = 1$, and $\nu = 0.002$ until $t = 2\pi$, one lap around the box. Run RK4 with a time step of 0.02 using Fourier derivatives, then using centred differences. Run forward Euler with a time step of 0.005 using Fourier derivatives. Plot the three final profiles with the exact solution as `week4/evidence/line-accuracy.png`. Print the maximum error of each run.

Use the library's Fourier derivatives to evolve the pulse with standard deviation 0.35 until $t = 1$ ($n = 64$, $c = 1$, $\nu = 0.05$). Run forward Euler, midpoint, RK4, and an RK4 whose four stage weights are equal, each at time steps 0.02, 0.01, 0.005, and 0.0025. Add a second panel to `week4/evidence/line-accuracy.png` showing maximum error against time step as points on log-log axes. Add a linear fit in log-log space for each method and label its slope in the legend.

Open `evidence/line-accuracy.png`.

Expected output: the figure below. Required: the RK4 pulse with Fourier derivatives overlaps the exact solution; the finite-difference pulse arrives late with ripples behind it; Euler's comes back taller; the slopes are within 15% of 1, 2 and 4, and the equal-weight RK4 falls to about 2. The answer key's slopes are 1.03, 2.01, 4.00, and 2.00 for the equal weights.



Pulse propagation and error convergence. Left: one lap of a pulse that contains a few dozen waves. Right: the error at $t = 1$ against the step for the three steppers and the equal-weight RK4, each with its fitted slope.

Explanation. The left panel compares errors from spatial differentiation and time integration. With the exact multipliers, RK4's small step gives a pulse that overlaps the exact solution; with centred differences the same stepper arrives late, because Equation 9's ik became $i \sin(k\Delta x)/\Delta x$ and the pulse's short waves fall behind the long ones, which is the wake. Euler at a step four times smaller still comes back taller: no piece of the imaginary axis is inside its region, so every travelling mode grows a little each step, the same growth that drifted week 2's energy. The right panel measures the accuracy order: at steps where all four runs are stable, the errors fall as h , h^2 and h^4 , and the equal-weight RK4 falls as h^2 , a hundred thousand times above RK4 at the smallest step. Its four stages cost four evaluations of F and buy a second-order answer; the weights 1, 2, 2, 1 are what makes RK4 fourth order. A slope near 2 for a line labelled RK4 is the signature of this mistake.

Part 2: Discretize the flow and build the solver

Goal: choose a periodic grid, compute spatial derivatives spectrally, and build the flow solver on the integrators of Part 1. Verify the derivative operators and the Taylor–Green decay with RK4: energy to six decimals and velocity error below 10^{-5} .

① DO

Build two tools: `field` writes a velocity field and `fluid` integrates it with the integrators of Part 1. Each command gets one design file specifying its arguments and its outputs: create `field.design.toml` and `fluid.design.toml` and copy each green panel into the file it names. The field generator first, with its two cases; the exact case takes a time, so the same tool writes the answer the solver is compared with:



`field.design.toml`

```
usage      = "field taylor-green --n 64 --nu 0.1 --t 1"
description = "write a velocity field to stdout; no unstated defaults"

[args]
n = "grid points per side"

[writes]
stdout = "one JSON object: case, n, seed, k_band, then u and v as n*n arrays"

[taylor-green]
usage      = "field taylor-green --n 64 --nu 0.1 --t 1"
description = "u = cos(x)*sin(y)*exp(-2*nu*t), v = -sin(x)*cos(y)*exp(-2*nu*t)"

[taylor-green.args]
t = "time of the exact solution; default 0, the initial field"
nu = "viscosity of the decay; required if t > 0"

[random]
usage      = "field random --n 128 --seed 2026 --k-min 2 --k-max 6"
description = "vorticity modes of equal amplitude, k-min <= |k| <= k-max, E(0) = 0.5"

[random.args]
seed = "seed of the phases, uniform in [0, 2*pi), drawn in an order independent of n"
k-min = "lowest |k| = sqrt(kx^2 + ky^2) kept, inclusive; modes outside are zero"
k-max = "highest |k| kept, inclusive"
```

```

fluid.design.toml
usage      = "field taylor-green --n 64 | fluid --method rk4 --nu 0.1 --dt 0.01 --t-
end 1 --every 0.1 --out artifacts/taylor-green"
description = "integrate the field on stdin with a chosen time integrator, record
frames; no unstated defaults"

[args]
method = "time integrator: euler, rk2, or rk4"
nu      = "kinematic viscosity"
dt      = "time step"
t-end   = "final integration time"

every = "time between snapshots; save at step 0 and every round(every/dt) steps after
that; a step is never shortened to land on a snapshot"
out    = "output folder"

[writes]
stdout = "a header, then one tab-separated line per snapshot: t, E (energy), Z
(enstrophy); the run stops at the first non-finite energy, prints that line, and exits
1"

"<out>/run.json"      = "case, n, seed, k_band as piped; method, nu, dt, t_end,
snapshot_every"
"<out>/fields.jsonl" = "per snapshot stored before the run stopped: t, step, u, v,
omega as n*n arrays, 6 decimals"

```

Each design file specifies one tool. `usage` gives an example command, and `description` states what the tool does. `[args]` explains the flags, and `[writes]` describes the terminal output and saved files. A table such as `[random]` specifies a subcommand with its own usage and flags. The `pipe` is the interface between the two tools: `field` writes one JSON object, `fluid` reads it and copies its metadata into `run.json`.

Build `field` and `fluid` in `week4/` in Rust, on the integrators and the `Integrator` trait already there, integrating the two-dimensional incompressible vorticity equation on a periodic $[0, 2\pi]^2$ grid: Fourier pseudospectral derivatives; the two-thirds rule, keeping only $|k_x|, |k_y| \leq \text{floor}(n/3)$ in the vorticity itself and in every product; the integrator chosen by `--method`, as `week4/field.design.toml` and `week4/fluid.design.toml` specify.

2 UNDERSTAND

Sample the periodic domain. The square $[0, 2\pi]^2$ is sampled at $x_j = j\Delta x$, $y_l = l\Delta x$, with $\Delta x = \frac{2\pi}{N}$ and $j, l = 0, \dots, N-1$. The endpoint 2π is omitted because it is the same location as 0. Each scalar field becomes N^2 numbers; the array entry $lN + j$ stores $f(x_j, y_l)$.

These samples determine a finite Fourier sum, Equation 7 applied along each index, with wavenumbers k_x, k_y from $-\frac{N}{2}$ to $\frac{N}{2} - 1$ in the FFT's order of Part 1. Its normalization must agree with the formulas: an unnormalized forward transform needs a factor $\frac{1}{N^2}$ in the inverse.

Compare local differences with Fourier differentiation. The centred differences of Equation 8 apply along each coordinate, with neighbours wrapping at the periodic edge: the first formula along x and then y gives a mixed derivative, and the second-derivative formulas added along both coordinates give the five-point Laplacian. Their errors fall as Δx^2 even for a single smooth wave; the Fourier representation offers a different choice for our periodic domain.

Compute spatial derivatives. Each derivative of a Fourier wave brings down its wavenumber. Repeated differentiation therefore uses repeated multipliers:

$$(\widehat{\partial_x^a \partial_y^b f})_{\mathbf{k}} = (ik_x)^a (ik_y)^b \hat{f}_{\mathbf{k}}. \quad (12)$$

Here $a + b$ is the *derivative order*. The gradient uses first derivatives, while the `Laplacian` sums second derivatives. The operators needed by the flow are:

Operator	Fourier multiplier	Role in the solver
$\partial_x f, \partial_y f$	ik_x, ik_y	Velocity, vorticity, and advection
$\partial_x^2 f, \partial_y^2 f$	$-k_x^2, -k_y^2$	Curvature in each direction
$\partial_x \partial_y f$	$-k_x k_y$	Commuting mixed derivatives in $\nabla \cdot \mathbf{u} = 0$
$\nabla^2 f$	$-(k_x^2 + k_y^2)$	Viscous diffusion and the Poisson equation

These multipliers differentiate every retained wave exactly. The approximation lies in representing the continuous field with finitely many waves. A smooth, periodic field is well suited to this representation: its omitted coefficients decay rapidly, whereas a fixed-width finite-difference stencil has a fixed power of Δx as its accuracy law. Walls or discontinuities would change this choice.

For a check with known derivatives, use $g(x, y) = \sin(3x) \cos(2y)$. Its derivatives are $\partial_x g = 3 \cos(3x) \cos(2y)$, $\partial_x^2 g = -9g$, $\partial_x \partial_y g = -6 \cos(3x) \sin(2y)$, and $\nabla^2 g = -13g$. All these waves fit on $N = 32$, so the derivative comparison tests the operators without a spatial truncation error.

The same multipliers turn the Poisson equation into one division per mode:

$$\hat{\psi}_{\mathbf{k}} = \frac{\hat{\omega}_{\mathbf{k}}}{k_x^2 + k_y^2}, \quad \hat{\psi}_{\mathbf{0}} = 0. \quad (13)$$

The zero mode fixes the otherwise arbitrary mean of ψ ; no velocity derivative sees it. Differentiating ψ then recovers the velocity for the next step.

Use each representation for the operation it simplifies. The Fourier coefficients make derivatives and the Poisson solve inexpensive. Their inverse transforms give velocity and gradient values at the grid points; transforming $ik_x \hat{\omega}$ back gives $\partial_x \omega(x_j, y_l)$ without a finite-difference stencil. These real-space values make the nonlinear product inexpensive too: the velocity and the gradients are transformed back to the grid, multiplied pointwise into $u \partial_x \omega + v \partial_y \omega$, transformed forward, dealiased, and joined by the diffusion term before the time step. The resulting **pseudospectral** method differentiates in Fourier space and multiplies on the grid. A product in Fourier space would instead require a convolution over pairs of modes. The FFT switches representations in order $N^2 \log N$ operations, making the grid product efficient. The grid is therefore used throughout the calculation, even though derivatives use Fourier multipliers.

Dealias the nonlinear product. Multiplying two grid fields produces waves with the sum of their input wavenumbers. These waves can be too short for the grid to distinguish from longer ones, an error called *aliasing*. The solver prevents their contribution to the retained modes by applying the **two-thirds rule**: it keeps only modes with $|k_x|, |k_y| \leq [N/3]$ in the fields and their products. This removal of aliased contributions is called *dealiasing* [2]. The highest wavenumber, $-N/2$, the *Nyquist* mode, is ambiguous for odd derivatives on an even grid; the cutoff removes it too. The solver applies the same cutoff to the vorticity it carries, not only to the products: modes beyond $N/3$ hold nothing, so no term of the equation ever acts on them.

Turn the PDE into time evolution. The spatial discretization leaves a vector $w(t)$ of Fourier coefficients. Its evolution is the system $dw/dt = F(w)$ of the opening section, Equation 4 with $y = w$, where one evaluation of F recovers velocity, computes advection, and adds diffusion. The integrators of Part 1 advance w using this F in Equation 10, without changes to the integrators themselves. At the intervals specified by `--every`, the tool stores a **snapshot**, a frame of the fields at that time. Each stage of midpoint and RK4 rebuilds velocity from its own vorticity; reusing the first stage's velocity throughout would integrate a different approximation and destroy the advertised order, as the equal weights did in Part 1.

Energy and enstrophy. The velocity field tells us how fast the fluid moves. We summarize its speed by the **kinetic energy** per unit mass, $E = \frac{1}{2} \langle u^2 + v^2 \rangle$, where the brackets mean an average over the box. The vorticity tells us its local rotation. We summarize the strength of that rotation by the **enstrophy**,

$$Z = \frac{1}{2} \langle \omega^2 \rangle. \quad (14)$$

Squaring the vorticity makes clockwise and counterclockwise rotation contribute positively instead of cancelling. This quantity measures rotation strength, not the number of vortices.

For a flow with the same speed amplitude, a shorter wavelength gives larger velocity derivatives. Those derivatives make the enstrophy larger even when the energy is unchanged. For one divergence-free Fourier mode, the contributions

obey $Z_k = |k|^2 E_k$. Energy measures squared speed; enstrophy measures squared vorticity and gives short wavelengths more weight. Doubling the wavenumber at fixed energy multiplies the enstrophy by four, which is why the short waves set the decay rates in Part 3.

The random flow. The random flow starts from vorticity modes of equal amplitude, the Fourier modes with $2 \leq |k| \leq 6$, each with a random phase drawn from a seed, scaled so that $E(0) = 0.5$: a sum of a few dozen waves, smooth, and different for every seed. Advection stretches its vorticity into thin filaments, which populate short waves; viscosity removes each mode at the rate $\nu|k|^2$, the short waves fastest. The flow loses much of its enstrophy while retaining much of its energy, because the disappearing short waves carry more enstrophy per unit energy.

The Taylor–Green solution and its zero advection. To test the solver, we use the Taylor–Green solution, a closed-form solution of Equation 3 found by Taylor and Green [3]:

$$u = \cos x \sin y e^{-2\nu t}, \quad v = -\sin x \cos y e^{-2\nu t}, \quad \omega = -2 \cos x \cos y e^{-2\nu t}. \quad (15)$$

Its energy is $E = \frac{1}{4}e^{-4\nu t}$ and its enstrophy is $Z = \frac{1}{2}e^{-4\nu t}$. Every part of the field decays at one rate and the vorticity pattern stays fixed. Its Fourier modes all have $|k|^2 = 2$ and form four counter-rotating cells, which the grid represents exactly at any $N \geq 4$. The relative error of a computed field f against an exact one is $\|f - f_{\text{exact}}\|/\|f_{\text{exact}}\|$, with $\|\cdot\|$ the square root of the sum of squares over the grid.

Here $\omega = 2\psi$, so $\nabla\omega$ is parallel to $\nabla\psi$, and $\mathbf{u}$ is perpendicular to $\nabla\psi$ by construction. The advection term $\mathbf{u} \cdot \nabla\omega$ is therefore identically zero. Taylor–Green tests the Poisson solve, derivatives, viscous term, and time stepper, but its zero advection term leaves nonlinear transport untested. That term is checked in Part 3, on a random flow, by eye, filaments forming and fading, and by its enstrophy decay.

- Read Boyd, *Chebyshev and Fourier Spectral Methods* [4], free, chapter 2, “Chebyshev and Fourier series”, sections 2.1–2.4. Find the statement that the error of a Fourier series of a smooth periodic function falls faster than any power of N .
- Read Trefethen [1], chapter 7, section 7.3, “Periodic grids”. Find the wavenumber range of the discrete Fourier transform and the sawtooth mode, the $N/2$ wave, whose derivative is set to zero.

KEY CONCEPTS

- A *periodic grid* samples the box without duplicating its endpoint; in two dimensions the multipliers of Part 1 act on each index.
- A *pseudospectral method* differentiates in Fourier space and multiplies fields on the grid; *dealiasing* keeps unresolved products out of the retained modes, and the solver carries no mode beyond the cutoff.
- The *streamfunction* turns the vorticity into a divergence-free velocity through one Poisson solve per stage.
- *Energy* measures squared speed; *enstrophy* measures squared vorticity and weights shorter waves more strongly.
- *Taylor–Green* has an exact decay rate, but its zero advection term leaves nonlinear transport untested.
- **The check:** derivatives of a known wave against their formulas, then the Taylor–Green decay against its exact energy and field.

3 VERIFY

- (1) **Represented waves have the correct derivatives.** Ask for a comparison using the solver’s own differentiation routines:



On the periodic $[0, 2\pi]^2$ grid with $n = 32$, apply the solver's Fourier differentiation routines to $g(x,y) = \sin(3x)\cos(2y)$. Compare dx , dxx , $dxdy$, and the Laplacian against their analytic derivatives. Also compute them with second-order centred finite differences on $n = 32$ and 64 , with periodic wrapping. Print maximum absolute errors and finite-difference error ratios in a table, using full-precision fields. Show me directly.

Read the printed comparison.

Expected output:

Derivative	Finite difference, $N = 32$	Finite difference, $N = 64$	Fourier, $N = 32$
$\partial_x g$	0.17050	0.04318	$< 10^{-10}$

$\partial_x^2 g$	0.25724	0.06487	$< 10^{-10}$
$\partial_x \partial_y g$	0.48534	0.12429	$< 10^{-10}$
$\nabla^2 g$	0.30838	0.07771	$< 10^{-10}$

Explanation. Halving the grid spacing reduces the finite-difference errors by about four, showing second-order accuracy for both first and second derivatives. Fourier differentiation reaches roundoff on these represented waves, so for them the only remaining approximation is in time, which the next check begins to test.

- (2) **The Taylor–Green field decays in place at the exact rate.** Install the tool, run the case, and read its final energy and enstrophy:

```
> cargo install --path . --quiet
mkdir -p evidence artifacts
field taylor-green --n 64 | fluid --method rk4 --nu 0.1 --dt 0.01 --t-end 1 \
  --every 0.1 --out artifacts/taylor-green > artifacts/taylor-green.tsv
head -2 artifacts/taylor-green.tsv; tail -1 artifacts/taylor-green.tsv
```

Expected output (six-decimal agreement):

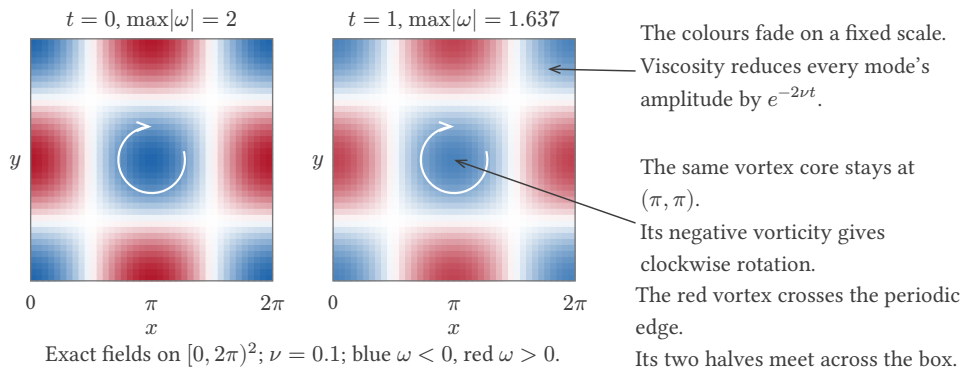
t	Energy E	Enstrophy Z
0.0	0.250000	0.500000
1.0	0.167580	0.335160

A wrong decay rate calls for a check of viscosity, wavenumbers, and the time step. The energy is one number and cannot tell a rotated or mirrored pattern from the right one, so write the exact field at $t = 1$ beside the recording and ask for the arrow-by-arrow comparison:

```
> field taylor-green --n 64 --nu 0.1 --t 1 > artifacts/taylor-green/exact-t1.json
```

Compare the last frame of `week4/artifacts/taylor-green/fields.jsonl` with the exact field `week4/artifacts/taylor-green/exact-t1.json`: the relative error of the velocity, printed in the terminal. Draw the vorticity at $t = 0$ and $t = 1$ with velocity arrows and a shared colour scale as `week4/evidence/taylor-green.png`.

Open `evidence/taylor-green.png`. **Expected output:** relative field error $\approx 7.0 \times 10^{-7}$, required $< 10^{-5}$, and the two panels below.



Taylor–Green decay, exact solution at $\nu = 0.1$. Compare positions and signs as well as amplitudes.

Explanation. The four alternating cells follow Equation 15. Their boundaries stay fixed because the velocity follows vorticity contours, making advection zero; their fading records viscous decay, the energy falling as the square of the amplitude. Stripes instead of cells suggest an error in the spatial derivatives; the field comparison catches a rotated or mirrored pattern the energy cannot.

Part 3: Find the stability limit

Goal: watch the random flow decay, its enstrophy falling by a factor of seven while its energy falls by less than half; predict RK4's diffusive and advective step limits from the stability function and see both in an energy plot of runs on either side of them; and tell that blow-up from the flow's own sensitivity, which grows a physical perturbation while the run stays stable.

1 DO

Run the seeded random field, the flow the next check and Part 4 use, then drive both cases past their limits. Start the recording in the second terminal:

```
> field random --n 128 --seed 2026 --k-min 2 --k-max 6 | fluid --method rk4 \
--nu 0.004 --dt 0.01 --t-end 10 --every 0.1 --out artifacts/random \
> artifacts/random.tsv
```

The run takes a few seconds. Then run Taylor–Green past its limit. The run stops when the energy stops being finite:

```
> mkdir -p artifacts/unstable
field taylor-green --n 64 | fluid --method rk4 --nu 0.1 --dt 0.04 \
--t-end 4 --every 0.1 --out artifacts/unstable/taylor-green \
> artifacts/unstable/taylor-green.tsv
```

Run Taylor–Green on a 64×64 grid with viscosity 0.1 until $t = 8$. Use RK4 with time steps 0.032 and 0.033. Run the random flow on a 128×128 grid with viscosity 0.004 until $t = 10$ (seed 2026, wavenumbers 2–6). Use RK4 with time steps 0.038 and 0.040, and Euler with a time step of 0.01. If both random RK4 runs blow up, try smaller time steps until one reaches $t = 10$. If both reach $t = 10$, try larger time steps until one blows up. Use the same random initial field and keep all other settings unchanged for these additional runs. Save snapshots every 0.5 time units for all runs. Save all runs under `week4/artifacts/scan/`. Print the largest speed of the random initial field. Plot energy against time with a logarithmic energy axis and one panel per case. Show both Taylor–Green runs. For the random flow, show one RK4 run that reaches $t = 10$, one that blows up, and the Euler run. Label each curve with its method and time step in the legend. Mark the stopping time of each unstable run. Save the figure as `week4/evidence/blowup.png`.

Run each case twice using RK4 with a time step of 0.01 until $t = 20$ (save snapshots every 0.5 time units). For the first run, use the original initial field. For the second run, add this ripple to the initial vorticity: $\delta \omega(x, y) = -7e-5 * M * \cos(3x) * \cos(4y)$. Here M is the largest absolute value among all initial u and v values of that case. Keep all other settings the same. Draw the distance between the two vorticity fields of each pair, relative to the size of the unperturbed one, against time on a log scale, as `week4/evidence/sensitivity.png`.

2 UNDERSTAND

The solver's spectrum against the stability regions. Part 1 drew the stability regions on the line's spectrum; the fluid's spectrum has the same two axes and many more points on them. Diffusion contributes $\lambda = -\nu|\mathbf{k}|^2$ for each retained mode: real and negative, largest for the corner mode $|\mathbf{k}|_{\max}^2 = 2\lfloor N/3 \rfloor^2$, the mode with $|k_x| = |k_y| = \lfloor N/3 \rfloor$ at the corner of the retained square, so the step must satisfy

$$\nu|\mathbf{k}|_{\max}^2 \Delta t \leq 2.785 \quad \text{for RK4,} \quad \leq 2 \quad \text{for Euler and midpoint.} \quad (16)$$

Advection carries a wave with wavevector $\mathbf{k}$ at the rate $i\mathbf{u} \cdot \mathbf{k}$ when the velocity is frozen: an imaginary λ . Its size is bounded by $U_{\max}|\mathbf{k}|_{\max}$, where $U_{\max}$ is the largest speed in the box and $|\mathbf{k}|_{\max} = \sqrt{2}\lfloor N/3 \rfloor$ the longest retained wavevector:

$$U_{\max}|\mathbf{k}|_{\max} \Delta t \leq 2.83 \quad \text{for RK4.} \quad (17)$$

For a purely oscillating mode, Euler and midpoint have no stable step at all: their growth per step is $|R|^2 = 1 + \theta^2$ and $1 + \theta^4/4$ for $z = i\theta$, fast for Euler and slow for midpoint. The solver therefore needs a region with both intervals, so the Taylor–Green and random-flow runs at $\Delta t = 0.01$ use RK4. As the grid is refined, the maximum stable step in Equation 16 shrinks as N^{-2} , while the time scale of the large-scale flow remains unchanged. The problem becomes *stiff*: rapidly decaying small-scale modes restrict the time step even when their amplitudes are negligible. Treating diffusion exactly, one factor $e^{-\nu|k|^2 h}$ per mode, removes Equation 16; the Challenge does so.

Predict the diffusive limit. At $N = 64$ the cutoff keeps $|k_x|, |k_y| \leq 21$, so $|k|_{\max}^2 = 2 \cdot 21^2 = 882$, and Equation 16 with $\nu = 0.1$ gives $\Delta t \leq 2.785/88.2 = 0.0316$. The Taylor–Green field itself lives at $|k|^2 = 2$, nowhere near this limit; the corner modes start at round-off, about 10^{-16} , and multiply by $|R(z)|$ every step. At $\Delta t = 0.033$ the factor is 1.2 and the growth needs some 200 steps to show; at 0.04 it is 2.8 and shows within 50. The stability boundary is sharp in $|R|$; the time to see it grows without bound as the step approaches the boundary from above. A solver whose diffusion acts on all N^2 modes, because it masks only the products, has $|k|_{\max}^2 = 2 \cdot 32^2 = 2048$ and blows up already near $\Delta t = 0.014$; measuring that value is a diagnosis of the solver, not a disagreement with the theory.

Predict the advective limit. For the random case at $N = 128$, the cutoff keeps $|k_x|, |k_y| \leq 42$, so $|k|_{\max} = 42\sqrt{2} = 59.4$; the initial field’s **largest speed** is $U_{\max} = 2.43$ for the answer key’s phases; another random generator draws other phases from the same seed, so your $U_{\max}$ differs and the bound follows it. Equation 17 gives $\Delta t \leq 2.83/(2.43 \cdot 59.4) = 0.0196$. This is a bound, not the boundary: it assumes the fastest fluid meets the shortest wave head on, and the flow also slows as it decays. The measured limit sits above the bound, within a small factor. The time step 0.01 is about half this bound for the reference initial field.

Sensitivity to a perturbation. A **perturbation** is a small change to the initial field. Here we add the vorticity ripple $\delta\omega = -7 \times 10^{-5} M \cos(3x) \cos(4y)$. The scale M is the largest absolute value among all initial u and v values of that case. The ripple has three periods along x and four along y . Both runs are valid solutions of Equation 3. The **distance** between them, $\|\omega_1 - \omega_2\|/\|\omega_1\|$, is a property of the flow, not of the integrator. In a flow with many interacting vortices that distance grows as $e^{t/\tau}$, with τ the *e-folding time*, until it is of order one: the flow is *sensitive* to its initial state, which is what *chaos* means here. In a laminar flow such as Taylor–Green the distance decays with the field. Neither run manufactures energy. A **blow-up** is the opposite thing: one run, round-off amplified by the stability function, energy created from nothing.

- Read Trefethen [1], chapter 4, “Accuracy, stability, and convergence”, the method-of-lines condition: the eigenvalues of the spatial operator, multiplied by Δt , must lie in the integrator’s stability region.

KEY CONCEPTS

- The *diffusive limit* Equation 16 follows from the corner mode of the retained spectrum; the *advective limit* Equation 17 from the largest speed and the longest wavevector, as a bound.
- A blow-up starts from round-off and needs steps to show: the boundary is sharp, the time to see it is not.
- *Sensitivity* can amplify the distance between two valid solutions while both lose energy through viscosity; numerical instability can amplify round-off and cause artificial energy growth.
- **The check:** the bracketing runs, Euler, and the perturbed pairs against the predictions, then the decay of the random run in four frames and two numbers.

3 VERIFY

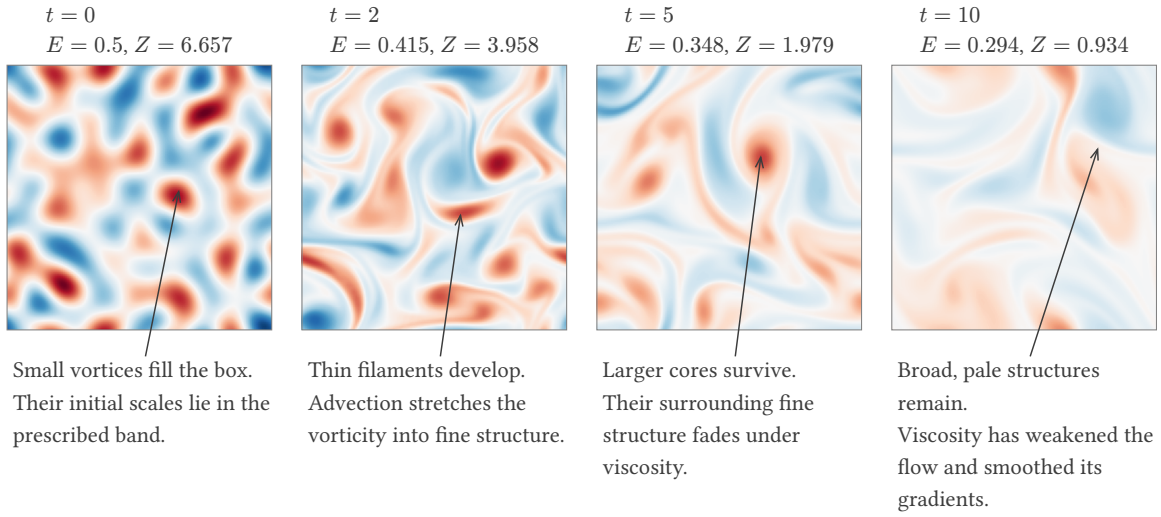
- (1) **Filaments fade first, vortices survive.** Read the first and last lines of the random-flow run at $\Delta t = 0.01$, then ask to see the flow:

```
> head -2 artifacts/random.tsv; tail -1 artifacts/random.tsv
```

Draw the vorticity of ``week4/artifacts/random/`` at $t = 0, 2, 5$, and 10 in one row with a shared colour scale, as ``week4/evidence/random.png``.

Open `evidence/random.png`.

Expected output: E falls from 0.50 to 0.29 while Z falls from 6.66 to 0.95, and the four frames below.



All frames use the same colour scale: blue -10.97 , white 0 , red 10.97 .

The frames show changing scales; the energy and enstrophy above each frame show the decay.

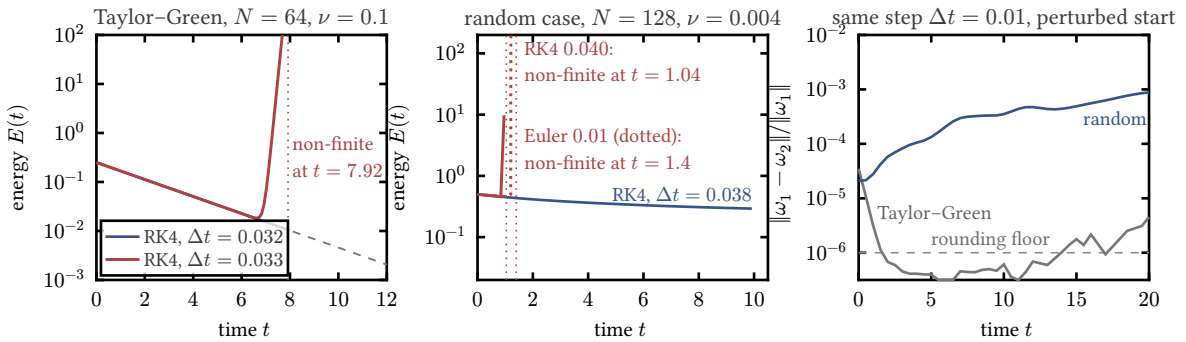
The random flow at four times. A fixed colour scale shows both the changing scales and the viscous decay: filaments form, thin, and fade, while the larger cores survive and merge.

Explanation. The frames and the two numbers say the same thing: the filaments are the short waves, they carry more enstrophy per unit energy, and viscosity removes them first, so enstrophy falls by a factor of seven while energy falls by less than half, kept by the large cores. An initial enstrophy far from 6.66 means the band or the amplitude is wrong: a square in k_x, k_y instead of a ring in $|k|$, or equal amplitudes on ψ instead of ω . The energy the flow loses is what viscosity dissipates, $dE/dt = -2\nu Z$, and a lower-order stepper dissipates it just as well, which is why Part 4 measures the order directly.

- (2) **The solver blows up where the stability function says.** Read the end of the run past the limit, then open `evidence/blowup.png` and `evidence/sensitivity.png`:

`> tail -2 artifacts/unstable/taylor-green.tsv`

Expected output: the last line is non-finite, before $t = 4$; the two plots follow, drawn here as one figure. A finite last line means the tool did not take the step it printed: a step shortened to land on a snapshot, a clamped step, or a filtered short wave.



Energy against time on either side of the two limits. Left, Taylor-Green at $N = 64, \nu = 0.1$: the run just below the diffusive limit follows the exact decay, the run just above departs from it and ends where the energy stops being finite. Right, the random case at $N = 128, \nu = 0.004$: RK4 on either side of its measured advective limit, and forward Euler at $\Delta t = 0.01$. The dashed line is the exact Taylor-Green energy. Third panel, the same step and a physical perturbation: the distance between two runs with a small initial perturbation grows in the random flow and decays to the rounding floor in Taylor-Green.

Required: the Taylor-Green boundary lies between 0.032 and 0.033 , within 5% of the predicted 0.0316 ; the random boundary lies between one and three times the bound of Equation 17 with your own largest speed (for the answer key, 0.0196 , and the boundary between 0.038 and 0.040). The random-flow panel shows one RK4 run

reaching $t = 10$ and one stopping early. Euler at 0.01 stops within the first two time units; the perturbed random pair drifts apart by more than tenfold by $t = 20$ while the Taylor–Green pair reaches the floor.

Left panel. Nothing in the Taylor–Green field drives the corner modes; round-off seeds them and the stability function amplifies them. The computed energy follows the exact decay until these errors become large enough to affect it; it then departs from the exact curve within a few steps. A boundary near 0.014 means the solver carries modes it was supposed to drop, as predicted.

Right panel. The measured boundary sits at about twice the bound, as the bound’s assumption allows, and the unstable run fails within the first time unit, while the flow is fastest. The time step 0.01 is about a quarter of the measured RK4 limit. Euler is unstable at this step: its energy grows from the first step and is non-finite by $t = 1.4$: the imaginary axis is outside its region.

Third panel. This blow-up is numerical, not turbulence. At a stable step the random flow amplifies a physical perturbation about forty-fold in twenty time units, exponentially at first with an e-folding time of about three, and both runs remain valid solutions; Taylor–Green damps the same perturbation a hundredfold until the two runs agree to six decimals. The difference between the two random-flow solutions grows while both lose energy through viscosity. Numerical instability instead amplifies round-off and produces artificial energy growth.

Part 4: Measure the accuracy order and choose a step

Goal: measure RK4’s accuracy order on the fluid from a step-refinement series, with a fitted slope within 15% of 4, then use the fourth-order law and Richardson estimation to choose the largest step of the random case whose temporal error is below 5×10^{-6} .

1 DO

Run Taylor–Green on an 8×8 grid with viscosity 0.5 until $t = 2$. Use RK4 with time steps 0.4, 0.25, and 0.2. Save each run under `week4/artifacts/order/rk4-dt<dt>/`. Compute the relative error of each final velocity field against the exact solution at $t = 2$. Plot the errors against the time step as points on log-log axes, add a linear fit in log-log space, and label its slope in the legend. Save as `week4/evidence/order.png`.

Run the random flow on a 128×128 grid with viscosity 0.004 until $t = 2$ (seed 2026, wavenumbers 2--6). Use RK4 with time steps 0.02, 0.0125, and 0.01. Compare these runs with a reference run using a time step of 0.0025 on the same grid. Report the relative error of ω at $t = 2$ for every run and the log-log slope of the series, as `week4/evidence/convergence.json`. Retain the final fields for error estimation.

The study takes about half a minute.

2 UNDERSTAND

Measuring accuracy order: grid and time steps. For the Taylor–Green run at $N = 64$, $\nu = 0.1$, $\Delta t = 0.01$, RK4’s temporal error on Taylor–Green is 2×10^{-14} : the recorded six decimals, a floor near 7×10^{-7} , hide it completely, and no refinement can show a slope there. The series therefore uses the coarsest grid, $N = 8$, a larger viscosity, $\nu = 0.5$, and steps at which $z = 2\nu\Delta t$ lies between 0.2 and 0.4: large enough that RK4’s error at $t = 2$ rises above the floor, small enough that it still follows its leading power h^4 , its *asymptotic range*. Every run stays inside the stability region, since $\nu|k|_{\max}^2\Delta t = 0.5 \cdot 8 \cdot 0.4 = 1.6 < 2.785$. An order measurement needs errors above the storage floor and steps inside the asymptotic range of the method.

Temporal error and its reference. The time stepper commits an error each step; for fourth-order Runge–Kutta the error after a fixed time is proportional to Δt^4 , so halving the step divides it by 16. The random case has no exact solution, so its reference is a finer run of the same code on the *same grid*, $N = 128$, at $\Delta t = 0.0025$: **self-convergence**. Every run starts from the same Fourier modes and phases; drawing a new field would compare different problems. The grid must be the same because the spatial error of $N = 128$, about 8×10^{-3} against a finer grid, buries every temporal error, which is below 3×10^{-5} . Self-convergence measures agreement with a finer run of the same model, not agreement with an exact solution; the same grid isolates the temporal error.

Fit the log-log slope as in Part 1 to measure the temporal accuracy order.

Richardson error estimation. The observed order also lets us estimate an error without knowing the exact solution. At fixed N , write the final field computed with step $h = \Delta t$ as $\omega_h = \omega_* + Ch^4 + o(h^4)$, where ω_* is the limit as the time step vanishes on this grid. The two fields computed with steps h and $2h$ then differ by approximately $(2^4 - 1)Ch^4$. Their difference gives the **Richardson estimate** of the finer run's relative temporal error:

$$e_h \approx \frac{\|\omega_{2h} - \omega_h\|}{(2^4 - 1)\|\omega_h\|}. \quad (18)$$

Here $\|\cdot\|$ is the norm of Part 2. The estimate **predicts the error at a nearby candidate step** h' as $e_{h'} \approx e_h(h'/h)^4$. This prediction is useful only where the measured order is close to four. The stored fields have six decimal places, so errors near the relative rounding floor, which scales with the field's amplitude and is about 1.5×10^{-7} in this study, can no longer reveal that scaling.

Choose a step. With this estimate, choose the largest candidate step whose predicted error at $t = 2$ is below 5×10^{-6} , then check the choice against the fine reference. The estimate concerns this grid and this final time; it is not a bound on the exact PDE solution. The final time is short on purpose. Part 3 showed that this flow amplifies any difference between two runs, so at long times two steps disagree at order one whatever the integrator's order.

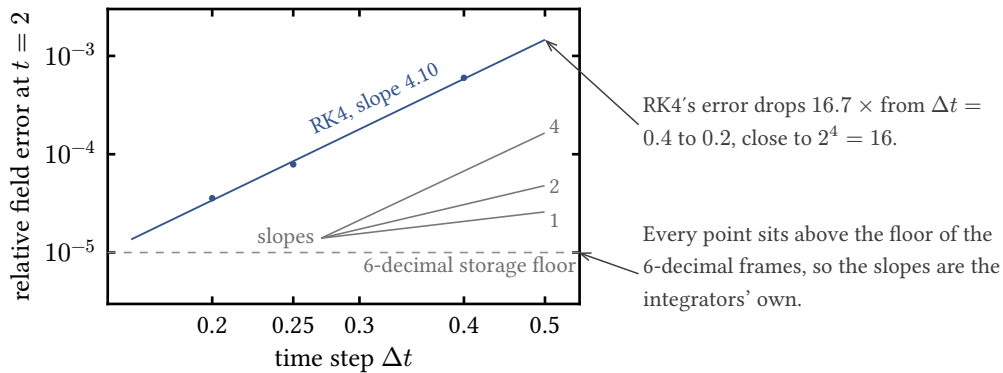
KEY CONCEPTS

- An *order measurement* needs errors above the storage floor and steps in the method's asymptotic range.
- *Self-convergence* compares a run with a finer run of the same code on the same grid, which isolates the temporal error.
- A *log-log slope* gives the power of the error law; fourth-order time stepping predicts a slope of four.
- *Richardson estimation* uses that order and two refined solutions to estimate the error and select a step.
- **The check:** RK4's slope against 4, then the random case's slope and the step it selects.

3 VERIFY

(1) **RK4's error falls with its advertised power on the fluid too.** Open `evidence/order.png`.

Expected output: the figure below, with slope 4.10. Check that the fitted slope is within 15% of 4.



Taylor–Green field error at $t = 2$ against the step, $N = 8$, $\nu = 0.5$. The dashed line is the six-decimal storage floor.

Explanation. The error falls along the fourth power, as on the line in Part 1, though the rate function is now nonlinear and the fields are stored to six decimals. RK4's smallest error sits only a few times above that floor, which is why the series stops at 0.2. A line labelled RK4 with slope near two is a stepper of the wrong order, which the decay of Part 3 cannot see.

(2) **The fourth-order law selects the step.** Read the errors of the random case and ask for the plot and the choice:

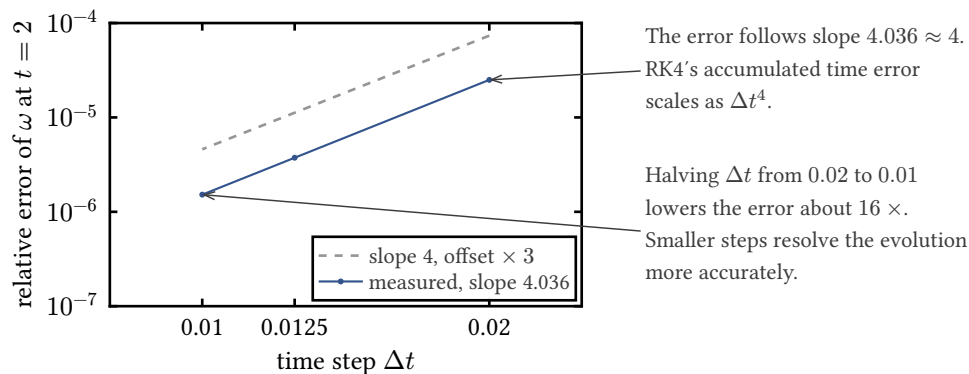
```
> cat evidence/convergence.json
```



Plot the errors in `week4/evidence/convergence.json` against time step as points on log-log axes. Add a linear fit in log-log space and label its slope in the legend. From the retained fields at $\Delta t = 0.02$ and 0.01 , estimate the relative temporal error at $\Delta t = 0.01$ using fourth-order Richardson estimation. Predict the errors at the candidate time steps 0.02 , 0.0125 , and 0.01 using fourth-order scaling. Choose the largest time step whose predicted error is below $5e-6$. Print the chosen step and its predicted and measured errors. Mark the choice on the plot. Save the figure as `week4/evidence/convergence.png`.

Open `evidence/convergence.png`.

Expected output: slope `4.024`, required $3.7 \leq q \leq 4.3$; the choice $\Delta t = 0.0125$, with predicted error 3.46×10^{-6} and measured 3.40×10^{-6} , both below 5×10^{-6} ; $\Delta t = 0.02$ fails.



Time-step refinement at $t = 2$. Every run uses $N = 128$; the reference uses $\Delta t = 0.0025$ on the same grid.

Explanation. The slope means that halving the step reduces the accumulated error sixteenfold, and the Richardson estimate turns two runs into a prediction the third run confirms. A flat curve suggests a mismatched reference grid or the rounding floor; a slope near one or two suggests a lower-order stepper. The estimate concerns this grid and $t = 2$; the physical checks of Part 3 remain necessary because both runs can share an error in the equation.

Wrap up the work

Goal: committed evidence for the physical claims and a README anyone can follow to regenerate it.

The committed tree in your exercise repository contains:

```
week4/
  Cargo.toml, Cargo.lock, field.design.toml, fluid.design.toml, README.md
  src/
  scripts/
  evidence/
    line-stability.png    ← measured stability map, a pulse on either side of the limit
    line-accuracy.png    ← pulse profiles and the four error slopes
    taylor-green.png     ← exact solution, vorticity at  $t = 0$  and  $1$ 
    blowup.png           ← energy on either side of both limits
    sensitivity.png       ← two runs a perturbation apart, both flows
    random.png           ← the random flow at  $t = 0, 2, 5, 10$ 
    order.png            ← RK4 on the fluid, slope 4
    convergence.png       ← step refinement and the chosen step
    convergence.json      ← the refinement errors and the slope
```

The `artifacts/` folders stay untracked. Keep them locally to regenerate the plots.

Compare the two design files you assembled with the reference `field.design.toml` and `fluid.design.toml`, block by block, to catch a copy error.

At the exercise repository root, send:



Write `week4/README.md` as a guide to regenerate every committed file from a clean clone: the install command, the two pipelines with their values, then each script with the file it writes, in order. List every file in `week4/evidence/` beside the command that produces it. Push to remote.

The crate is a tutorial, not a research code: it omits walls, three dimensions, the pressure solve of projection methods, and every treatment of stiffness. Research codes such as Dedalus [5] and spectralDNS [6] carry those; the Challenge takes up the stiffness thread.

Challenge: long-time vortex statistics in an inviscid fluid

Which long-time vortex statistics reflect the inviscid fluid, and which reflect the numerical treatment of small scales? Reproduce established methods first, then investigate this unresolved question for the week's periodic square. Modin and Viviani [7] already studied related questions on the sphere. How faithfully a finite matrix model captures long-time Euler dynamics remains a research problem [8]. Conservation checks establish that a method works as designed; the research claim needs evidence about the evolving flow.

Warm-up: exponential time stepping. Equation 16 exists because RK4 treats diffusion as one more term of F . Diffusion alone is solved exactly, one factor $e^{-\nu|\mathbf{k}|^2 h}$ per mode. An *integrating factor* applies that factor and lets RK4 handle advection only; the exponential time differencing scheme ETDRK4 [9], in the stable form of Kassam and Trefethen [10], also treats the linear term exactly. Reproduce the Taylor–Green decay at a step above RK4's diffusive limit, checking the error against the exact solution. Use the random flow to measure the remaining advective restriction. This is preparation for runs with damping; at $\nu = 0$ the diffusive restriction is already absent.

Preparation: a structure-preserving integrator. The inviscid vorticity equation conserves energy and every spatial moment of vorticity. These moments are examples of *Casimirs*, invariants of its Hamiltonian structure. Zeitlin's truncation [11] approximates this equation by a flow of matrices. Its vorticity matrix W keeps its eigenvalues, so the flow is called *isospectral*. The isospectral midpoint method [7] preserves those eigenvalues in exact arithmetic. Its discrete Casimirs are matrix traces $C_m = \text{tr}(W^m)$. With the truncation's normalization, these approximate continuum vorticity moments as the resolution increases.

Implement the periodic version and compare midpoint with RK4 on the **same matrix equation**. Plot relative changes in C_2 , C_4 , and energy in `evidence/invariants.png`, using full-precision diagnostics. Tighten the implicit solve tolerance and halve the time step separately to distinguish solve error from time-stepping error. Report the largest drift over the run and how it changes under refinement. This reproduces a known method; its energy error still needs measurement.

Investigation: energy in the largest scales. Use the fraction of kinetic energy in Fourier modes $0 < |\mathbf{k}| \leq 2$ as a measurable statistic of the large-scale flow. Measure it from each method's reconstructed velocity field. Compare the matrix method with the week's Fourier solver at $\nu = 0$. Add a Fourier run with *hyperviscosity*, artificial damping concentrated at large wavenumbers, by adding $-\mu|\mathbf{k}|^4 \hat{\omega}$ to each mode's rate, with $\mu > 0$. Measure the energy and enstrophy this damping removes.

For each seed, project the same initial physical field into each method and check its initial energy and enstrophy. Choose the averaging interval before comparing the results. Repeat across seeds, halve the time step, increase spatial resolution, and vary the damping strength. Compare methods at matched resolved wavelengths; matrix size and Fourier grid size alone do not establish that match. Also extend the averaging interval to check for continuing evolution.

Save the per-seed statistics and run parameters in `evidence/vortex-statistics.csv`. Plot the mean energy fraction with uncertainty across seeds in `evidence/vortex-statistics.png`, showing the refinement and damping comparisons. Success is a supported conclusion about this observable over the tested interval: a persistent difference, agreement within a quantified uncertainty, or unresolved dependence on resolution or duration. A difference between spatial discretizations alone cannot identify Casimir conservation as its cause. Before claiming a new result, establish what the literature already shows for the chosen geometry and initial conditions.

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